DSP Special Assignment

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May 19, 2026

Leslie / Rotary Loudspeaker Effect

What the Effect Does

The Leslie effect, also called the rotary loudspeaker effect, simulates the sound of a loudspeaker cabinet whose horn rotates while playing. The result is a lively moving sound: the signal seems to swirl across the stereo image, while its loudness and pitch change periodically. At moderate rotation speeds the effect adds motion and depth, especially to organ-like sustained sounds. At higher speeds it becomes more dramatic, with a clear tremolo-like amplitude variation and a vibrato-like Doppler shimmer.

The effect is strongly associated with electric organs, but it can also be used on guitar, voice, synthesizers, and other sustained sounds. The important audible ingredients are not only a left-right stereo movement, but also small periodic pitch and intensity changes. These changes make the sound feel as if it is physically moving relative to the listener.

Artistically, the Leslie speaker became important because it made electric organs sound less static and more expressive, closer in spirit to the moving air and spatial depth of a pipe organ. In the 1960s musicians also began using it as a studio effect beyond organ sounds. The Beatles, for example, used Leslie processing on guitar and vocals, including George Harrison's guitar parts.

How the Effect Works

The implementation follows the rotary loudspeaker model described in DAFX, Section 3.4.3. A Leslie cabinet is approximated by two opposite rotating horns. When a horn moves toward the listener, the perceived pitch and level increase; when it moves away, the perceived pitch and level decrease. The opposite horn performs the complementary motion, so the two horns are controlled by signals with a 180° phase difference.

The effect is implemented using two modulated fractional delay lines. Changing the delay over time produces a Doppler-like pitch modulation, similar to vibrato. The outputs of the delay lines are then amplitude modulated using the same low-frequency oscillator, which models the directional intensity change of the rotating horn. Finally, the two horn signals are mixed unequally into the left and right channels to create a stereo rotary-speaker impression.

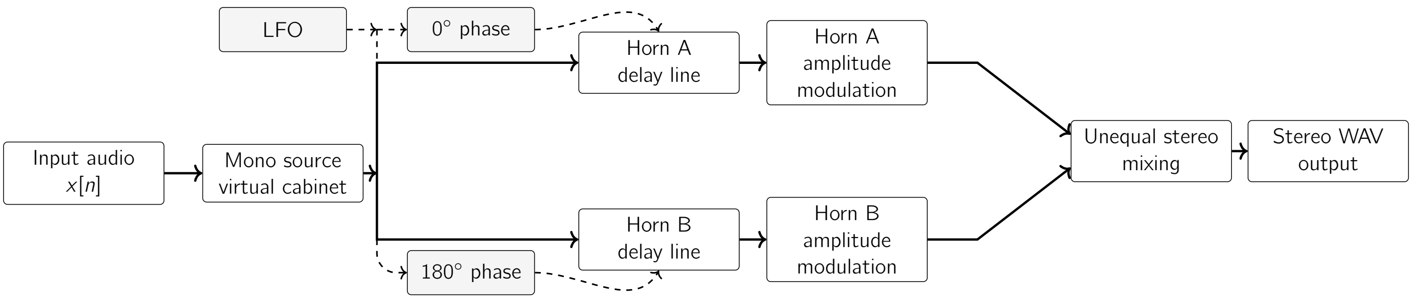

Figure 1: Signal flow of the Leslie effect implementation.

Let the low-frequency oscillator be

$$m[n] = \sin\left(2\pi f_{\mathrm{rot}} \frac{n}{f_s}\right)$$

The two opposite horn delays are

$$D_A[n] = D_0 + \Delta D \cdot m[n]$$
$$D_B[n] = D_0 - \Delta D \cdot m[n]$$

The delayed horn signals are

$$h_A[n] = x\left[n - D_A[n]\right]$$
$$h_B[n] = x\left[n - D_B[n]\right]$$

The synchronous amplitude modulation is

$$a_A[n] = 1 - \alpha \cdot m[n]$$
$$a_B[n] = 1 + \alpha \cdot m[n]$$

Thus,

$$s_A[n] = a_A[n] \cdot h_A[n]$$
$$s_B[n] = a_B[n] \cdot h_B[n]$$

The stereo output is produced by unequal mixing:

$$y_L[n] = s_A[n] + \beta \cdot s_B[n]$$
$$y_R[n] = \beta \cdot s_A[n] + s_B[n]$$

Here, f_{rot} is the rotation frequency, D_0 is the center delay, ΔD is the delay modulation depth, α is the amplitude modulation depth, and β controls the stereo cross-mix.

Implementation Choices

The program is written in Python using NumPy and SciPy. It accepts an input WAV file and writes a stereo processed WAV file:

```
python3 leslie.py dry/input.wav wet/output_leslie.wav
```

The default settings are chosen to make the effect clear on a short demonstration sample:

- rotation rate: 6 Hz

- center delay: 5 ms
- delay depth: 2 ms
- amplitude depth: 0.55
- stereo cross-mix: 0.7

For the final sound demonstrations, I produced three versions of the given dry samples. I chose three different styles because the Leslie effect can be subtle or very obvious depending on the rotation speed, Doppler depth, amplitude modulation depth, and stereo separation. Using three parameter sets makes the perceptual change easier to hear and also demonstrates how the same algorithm behaves on different types of audio material.

Table 1: Leslie effect demonstration parameter sets

Style	f_{rot}	D_0	ΔD	α	β	Purpose
Slow wide	2.0 Hz	6 ms	1.0 ms	0.35	0.6	Slower and smoother motion; useful for hearing the spatial movement without too much pitch wobble.
Classic fast	6.5 Hz	6 ms	3.0 ms	0.85	0.4	A clear fast Leslie setting with strong tremolo and Doppler modulation.
Dramatic	8.5 Hz	9 ms	5.0 ms	0.95	0.2	An exaggerated setting chosen to make the rotary character very obvious.

The parameter β was reduced in the stronger examples so that the left and right channels receive more different mixtures of the two horns. This makes the stereo rotation more noticeable. The larger ΔD values increase the Doppler-like pitch motion, and the larger α values increase the tremolo-like intensity modulation.

The code is sample-rate independent: millisecond parameters are converted to samples using the input file's sampling rate. The input is summed to mono before processing, because the model represents one source being played through a virtual rotating cabinet. The output is stereo. After processing, the signal is normalized if necessary to avoid clipping.

A more detailed Leslie model could split the signal into high-frequency horn and low-frequency drum bands, use separate rotation speeds, add acceleration between slow and fast modes, or model cabinet coloration and room reflections. Here, the main goal is to demonstrate the core DSP ideas from DAFX: delay-line modulation, amplitude modulation, opposite-phase horn motion, and stereo mixing.

References

U. Zolzer, ed., *DAFX: Digital Audio Effects*, 2nd edition. section 3.4.3 for the rotary loudspeaker effect, and Chapter 2 for fractional delay lines and vibrato based on modulated delay.